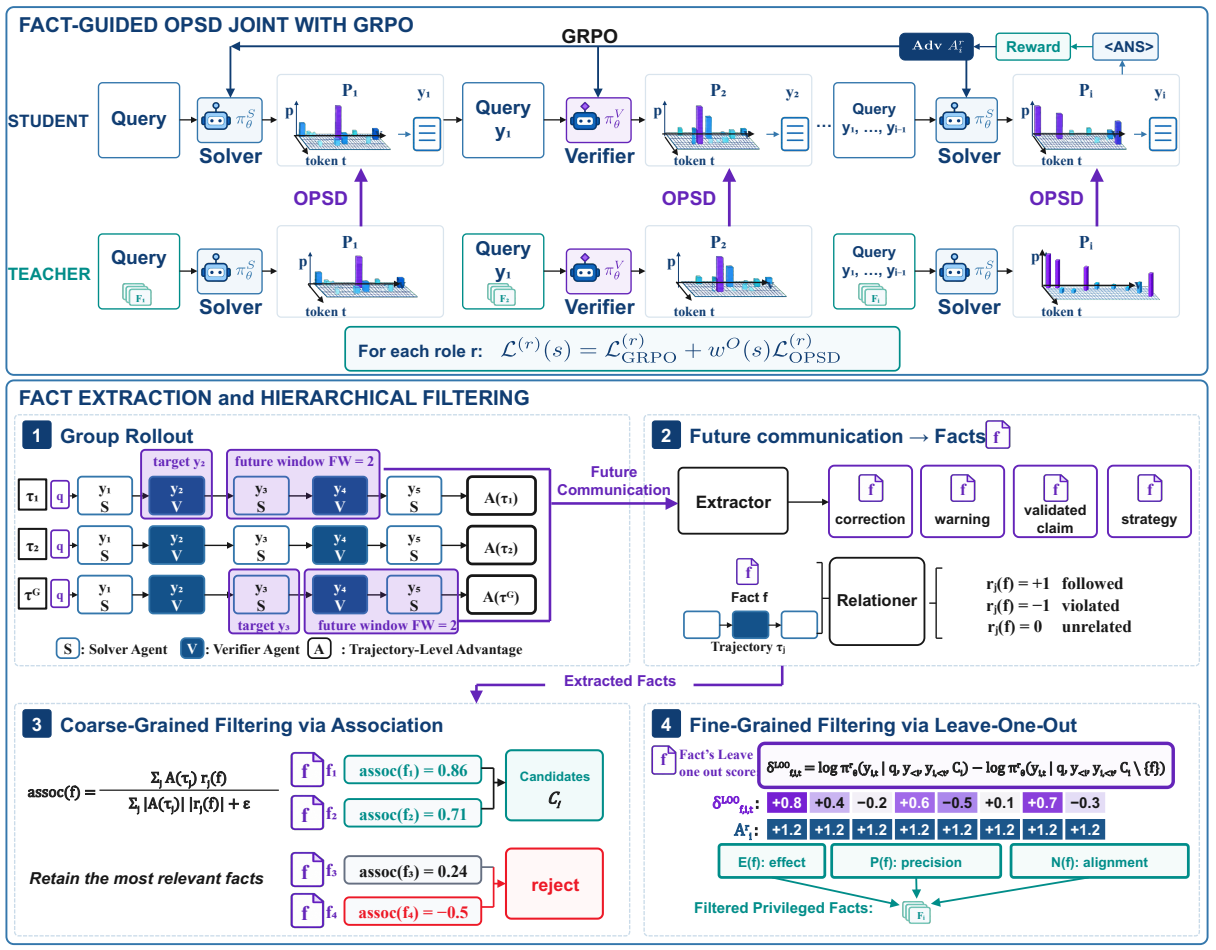

FACT-GUIDED OPSD JOINT WITH GRPO
GRPO
Adv A_i^r
Reward
<ANS>
STUDENT
TEACHER
Query
Solver
Verifier
OPSD
token t
For each role r: $\mathcal{L}^{(r)}(s) = \mathcal{L}^{(r)}_{\text{GRPO}} + w^O(s)\mathcal{L}^{(r)}_{\text{OPSD}}$
FACT EXTRACTION and HIERARCHICAL FILTERING
1 Group Rollout
target y_2
future window FW = 2
target y_3
$A(\tau_1)$
$A(\tau_2)$
$A(\tau^G)$
S: Solver Agent
V: Verifier Agent
A: Trajectory-Level Advantage
Future Communication
2 Future communication → Facts
Extractor
correction
warning
validated claim
strategy
Fact f
Trajectory τ_j
Relationer
$r_j(f) = +1$ followed
$r_j(f) = -1$ violated
$r_j(f) = 0$ unrelated
Extracted Facts
3 Coarse-Grained Filtering via Association
$\text{assoc}(f) = \frac{\Sigma_j A(\tau_j)\, r_j(f)}{\Sigma_j |A(\tau_j)|\, |r_j(f)| + \varepsilon}$
f_1 assoc(f_1) = 0.86
f_2 assoc(f_2) = 0.71
Candidates C_t
Retain the most relevant facts
f_3 assoc(f_3) = 0.24
f_4 assoc(f_4) = −0.5
reject
4 Fine-Grained Filtering via Leave-One-Out
Fact's Leave one out score
$\delta^{\text{LOO}}_{i,t} = \log \pi^s_\theta(y_{i,t} \mid q, y_{<t}, y_{i,<t}, C_t) - \log \pi^s_\theta(y_{i,t} \mid q, y_{<t}, y_{i,<t}, C_t \setminus \{f\})$
$\delta^{\text{LOO}}_{i,t}$: +0.8 +0.4 −0.2 +0.6 −0.5 +0.1 +0.7 −0.3
A^r_t: +1.2 +1.2 +1.2 +1.2 +1.2 +1.2 +1.2 +1.2
E(f): effect
P(f): precision
N(f): alignment
Filtered Privileged Facts: